

Unit

2

Lesson

2



Innovators in Action.

Light-Up Swings for Everyone.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Design and create a swing that is wide and safe for children who use wheelchairs.
- Build a simple electric circuit using a battery, wires, and a switch to power LED lights and sound for their swing.
- Explain how light and sound can help make playground swings more fun and easier to use.
- Describe how the swing moves using words like fast, slow, up, and down.
- Differentiate between push button and On/off switch.

Challenge questions of the lesson

1. How can you put the lights on your swing so everyone can see them, even when the swing is moving?
2. Why might having lights and sounds on a swing make it more fun for different kids to play together?
3. If you could make the lights or sounds do something special when the swing moves, what would it be?

SEL Relation

- **Creative Expression (Self-Awareness):** Students will use their creativity to design and implement the light and sound features on their swings, expressing their ideas and understanding of how these elements can enhance play.
- **Empathy and Inclusive Design (Social Awareness):** Students will consider how the added sensory elements can make the swing more engaging and accessible for peers with diverse sensory needs, fostering empathy and understanding.
- **Problem-Solving and Perseverance (Self-Management):** Students may encounter challenges in integrating the circuits into their models, requiring them to problem-solve and persevere to achieve a functional outcome.
- **Communication and Collaboration (Relationship Skills):** If working in groups, students will need to communicate their ideas and collaborate to decide on the best way to integrate the lights and sounds into their accessible swing design.

Engage



Teacher Role:

- **Facilitator:** Guide the discussion about the story, highlighting the students' creative ideas for the swing.
- **Questioner:** Pose questions to encourage students to think about how they could make the swing light up and change colors.
- **Connector:** Help students connect the idea of lights to things they already know and understand (e.g., flashlights, toys with lights).
- **Enthusiast:** Show excitement about the students' imaginative ideas to foster a positive and creative learning environment.
- **Guide:** Steer the conversation towards the practical aspects of making the light-up swing, hinting at the need to understand how lights work.

Expected From Students:

- **Active Listening:** Students will listen attentively to the story and engage with the characters' brainstorming.
- **Creative Thinking:** Students will be inspired to think creatively about how to make a swing more fun and visually engaging.
- **Idea Sharing:** Students will share their own ideas about how to make things light up or change colors.
- **Initial Problem-Solving:** Students will begin to think about the basic requirements for making lights work on a swing.
- **Curiosity:** Students will become curious about how LEDs work and how they can be connected.

Parts of the books included.



Story



- Set the Scene with Enthusiasm

Begin by saying:

“Class, today we’re going to make something amazing—a swing that lights up when you play on it!”

Then ask:

“Why do you think lights might be helpful or fun on a swing?”

- Hook with the Story Dialogue

Introduce the story:

“Let’s see what Adam, Laila, and Omar came up with when they started talking to Miss Sarah about their idea!”

Read the story aloud using different voices for each character to bring energy and excitement.

Possible Strategies for Implementation:

- Role Play the Dialogue

Let students take on the parts of Laila, Adam, Omar, and Miss Sarah. This builds engagement and helps them connect with the idea of designing for a purpose.

- Brainstorm Together

After reading the story, ask:

“What do we already know about things that light up?”

“Where do we see LED lights in real life?”

Write their responses on the board.

- Visual Thinking Strategy

Show real LED strips, small lights, or videos of light-up toys to spark ideas. Let students observe and describe what they see and how they might use these in a swing design.

- Use Think-Pair-Share

Ask students:

“If you could make your own swing that lights up, what would it look like?”

Let them first think, then share with a partner, and finally discuss with the class.

- Connect to Real-World Purpose

Say:

“These lights can help kids who can’t see well know the swing is moving. We’re not just making it fun—we’re making it helpful for everyone!”

Let's Think!



Objective: Students will explore how sensory supports like lights and sounds can help children with disabilities use playground equipment more safely and confidently. This activity fosters empathy, inclusive thinking, and basic understanding of assistive technology features in playground design.

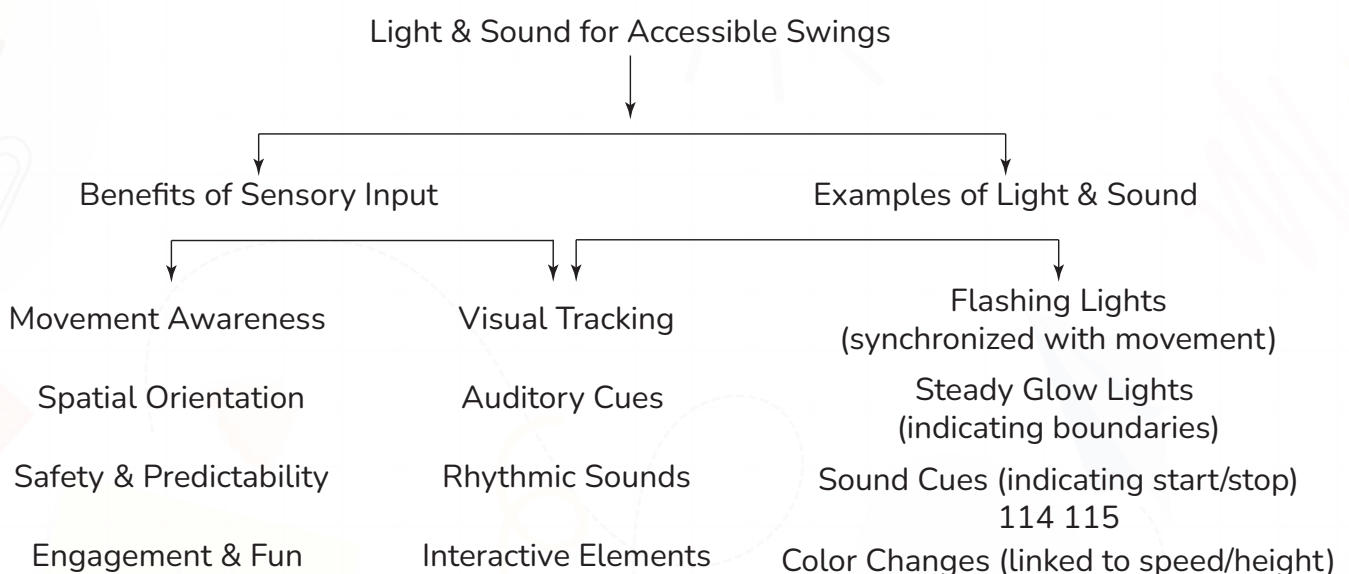
Possible Answers

How can light and sound help children with disabilities use a swing?

- ☒ Help them know when the swing is moving (Provides sensory cues for awareness and anticipation)
- ☐ Make the swing look nice only (Aesthetic appeal is secondary to functional benefit)
- ☒ Help them feel safe when playing (Predictable sensory input can reduce anxiety)
- ☐ Make the swing loud for fun only (Sound needs to be purposeful, not just loud)

Possible Strategies for Implementation:

- Think-Pair-Share (Focus on Sensory Experiences):
 - How to Implement: Ask students: "Think about times when you use your senses (sight and hearing) to understand what's happening around you. How do lights or sounds sometimes tell you something important?" (Think time). "Share your ideas with a partner." (Pair). "What are some ways light and sound could help someone know what a swing is doing?" (Share with the class, leading into the first question).
 - Benefit: Connects the abstract concept to students' own sensory experiences and highlights the informative nature of light and sound.
 - Brainstorm Web:



Explorer's Toolkit



Teacher Role:

- **Introducer:** Introduce the activity as an exploration of two simple circuits: one that makes a light turn on and off, and another that makes a sound. Explain that they will be using the smart pad to build these circuits.
- **Safety Guide:** Emphasize the importance of using the smart pad correctly and following instructions to avoid any issues.
- **Facilitator:** Guide students through the process of identifying the components needed for each circuit (light, switch, wire for the light circuit; buzzer, push button, wire for the sound circuit) and connecting them on the smart pad.
- **Observer:** Observe students as they build their circuits and test them, noting their ability to follow instructions and their understanding of how the switch and button control the flow of electricity.
- **Questioner:** Encourage students to describe what they observe with each circuit and to think about how the switch and button are different in how they control the light and sound.

Expected From Students:

- **Hands-on Participation:** Students will actively participate in building both the light circuit and the sound circuit on the smart pad.
- **Following Instructions:** Students will follow the instructions to connect the components correctly for each circuit.
- **Observation:** Students will carefully observe what happens when they flip the on/off switch in the light circuit and when they press the push button in the sound circuit.
- **Understanding Basic Circuit Control:** Students will begin to understand how a switch can turn a circuit on and off, and how a button can activate a component within a circuit.

Parts of the books included.

Explorer's Toolkit



DIY Lab

Let's Play with Circuits!

You're going to build two fun circuits using your smartpad:

Light Circuit:

Use a light, on/off switch, and a wire.

Watch what happens when you flip the switch!

Sound Circuit:

Use a buzzer, a push button, and a wire.

Try pressing the button — what do you hear?

What did you discover?

Choose what's true:

- ☐ The light stays on when I flip the on/off switch.
- ☐ The buzzer makes noise only when I press the button.
- ☐ Both buttons work the same way.
- ☐ The on/off switch doesn't work with the buzzer.
- ☐ Buttons help me start and stop things.
- ☐ I don't need wires to make it work.

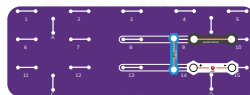


Figure 27: LED circuit with a push button.

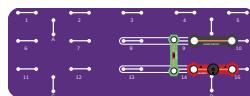


Figure 28: Buzzer circuit with a push button.



DIY Lab

Objective: Students will build and compare two types of basic circuits—one using a light and an on/off switch, and the other using a buzzer and a push button—using a smartpad. This activity helps students explore how switches control electricity and understand the difference between continuous and momentary circuits through hands-on investigation.

Materials Needed:

- Smartpad – magnet – power source
- LED light component
- Buzzer component
- On/off switch
- Push button
- Wires

Possible answers:

- ☒ The light stays on when I flip the on/off switch.
- ☒ The buzzer makes noise only when I press the button.
- ☐ Both buttons work the same way.
- ☐ The on/off switch doesn't work with the buzzer.

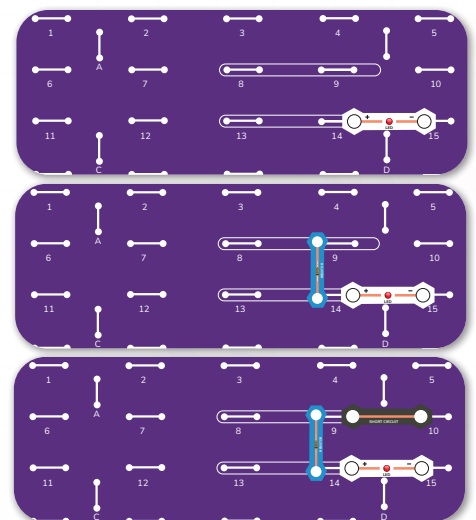
Explanation:

- The light circuit uses an on/off switch, so the light stays on once the switch is flipped.
- The buzzer circuit uses a push button, so the buzzer only activates while the button is pressed.
- Therefore, the two switches do not work the same way, and the on/off switch is not used with the buzzer in this setup.

Activity Instructions:

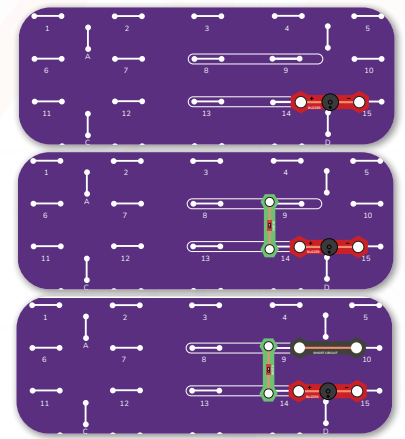
• LED Circuit

1. Connect the battery's positive end with the LED's positive end (the taller leg).
2. Connect the LED's negative end (the shorter leg) to a resistor's terminal.
3. Connect the other terminal of the resistor to the battery's negative end.
4. Connect smart pad with your power source and see what happens.



- **Buzzer circuit**

1. Connect the negative terminal of the battery to the negative terminal of the buzzer.
2. Connect the positive terminal of the buzzer to a terminal of the push button wire.
3. Connect the other terminal of the push button to the positive terminal of the battery with a wire.
4. Connect the smart pad with the power source.



Possible Strategies for Implementation:

- **Model First:**

Demonstrate both circuits in front of the class before students try it themselves to model how each switch functions.

- **Pair Collaboration:**

Let students work in pairs—one handles components while the other reads instructions and records observations.

- **Visual Checklists:**

Use visual aids or icons next to the steps and components to support emerging readers or ELL students.

- **Compare & Contrast Discussion:**

Facilitate a short group discussion where students compare the behavior of both switches using guiding questions like:

- “What’s the difference between flipping and pressing?”
- “Which switch would be better for a light or a buzzer?”



Teacher Role:

- **Observation Focus:** Guide students on what to look for in the video, such as the size of the swing seat, how children get onto the swing, any safety features, and how the swing moves.
- **Inclusion Advocate:** Emphasize the importance of designing playground equipment that allows everyone to play together.
- **Discussion Facilitator:** Lead a post-viewing discussion to explore the different accessibility features shown and encourage students to think about how these ideas could be incorporated into their own swing designs.

Expected From Students:

- **Attentive Viewing:** Students will watch the video carefully to observe the different

designs of accessible swings.

- **Feature Identification:** Students will identify and mentally note (or jot down) specific features that make the swings accessible to children with various needs.
- **Comparative Analysis:** Students will begin to compare the different swing designs and think about the advantages of each.
- **Active Participation:** Students will share their observations and ideas during the pre- and post-watching discussions.

Parts of the books included.



Screen Lessons

Let's watch a video together!

We'll learn how to make a swing that all kids can use.

<https://www.tiktok.com/@therecessproject/video/7355241063914818846>



Screen Lessons

Objective: Students will explore how accessible swing designs can support inclusive play by observing real-world examples of playground equipment that uses adaptive seating and support structures. By watching the video, students will understand how thoughtful engineering and design help children with disabilities participate safely and joyfully in physical play activities like swinging.

Video link: <https://www.tiktok.com/@therecessproject/video/7355241063914818846>



Possible Strategies for implementation:

1. Before Watching (Set the Purpose):

1. Prompt Curiosity with a Question:

- Ask: "Have you ever seen a swing that works for kids who use wheelchairs?"
- Let students share what they've seen or imagine what it might look like.

2. Activate Background Knowledge:

- Briefly review what "inclusive" means and how playgrounds can be made fair for everyone.

3. Introduce a Focus:

- Tell students to look for how the swing moves, who can use it, and what makes it special while watching.

2. During Watching (Active Engagement):

1. Visual Cues & Note-Taking:

- Ask students to jot or draw what they notice about how the swing works.

2. Pause & Discuss:

- Pause at key moments to ask guiding questions like:
 - “What do you see that helps the child feel safe?”
 - “What makes this swing different from the one you know?”

3. After Watching (Reflection & Application):

1. Think-Pair-Share:

- Students reflect individually, then discuss with a partner:
 - “What did you learn about accessible swings?”
 - “What could we add to our playground to make it more inclusive?”

2. Design Challenge Link:

- Prompt students to sketch or build a swing model using what they learned from the video (e.g., safety belts, ramps, wide space).

3. Connection to Career Readiness:

- Discuss who might design such swings (e.g., engineers, accessibility specialists) to show relevance to real-world careers.

Elaborate



Teacher Role:

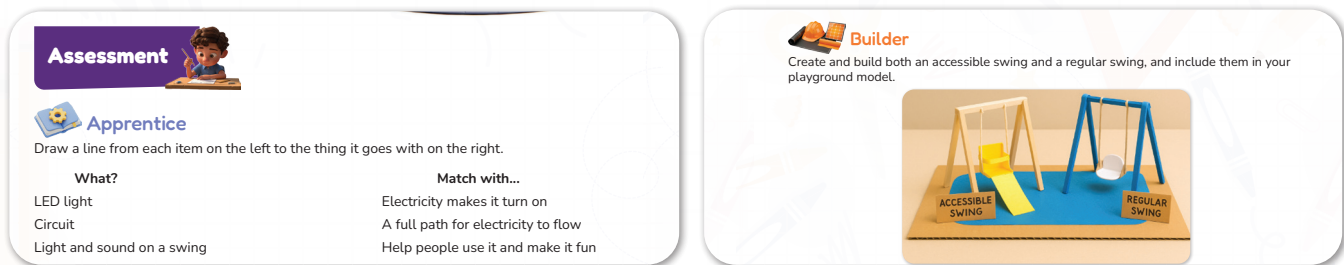
- **Facilitator:** Guide students through both tasks, ensuring they understand the concepts of accessibility and circuits in playground equipment.
- **Clarifier:** Explain what an accessible swing is and how it differs from a traditional swing.
- **Observer:** Monitor student engagement and identify those who may need support in matching circuit concepts or assembling physical models.
- **Encourager:** Reinforce creativity and problem-solving during the construction of both types of swings and correct vocabulary usage in describing how a circuit works.
- **Questioner:** Prompt students to explain their thinking with questions like:
 - “Why is it important for all kids to have a place to swing?”
 - “How does the LED light work in a circuit?”
 - “What did you learn about electricity when building this?”

Expected From Students:

- **Understanding of Basic Electrical Concepts:**
 - Correctly match each component (LED light and circuit) with its function:
 1. LED light → Electricity makes it turn on.
 2. Circuit → A full path for electricity to flow.

- **Application of Inclusive Design:**
 - Design and build two swings:
 1. One traditional swing.
 2. One accessible swing (e.g., wide seat, safety support, ramp).
- **Integration into a Playground Model:**
 - Incorporate both swing types into their overall playground model.
 - Demonstrate understanding of accessibility needs through design choices.
- **Communication:**
 - Clearly describe what they built and how it meets different children's needs.
 - Explain how electricity is used in their designs if applicable (e.g., LED-lit swings).

Parts of the books included.



Apprentice

Objective: Students will identify the basic function of an LED light and a circuit by correctly matching each with its purpose.

This supports foundational understanding of electrical systems and builds vocabulary related to electricity and how components work together.

Answers:

LED light → ☒ Electricity makes it turn on

Circuit → ☒ A full path for electricity to flow

Possible Strategies for Implementation:

1. Real-Life Demonstration First

- Use a working circuit with a battery, wires, and an LED light to show how electricity flows.
- Let students observe what happens when the circuit is complete or broken.

2. Think-Pair-Share

- Ask students: "What happens when you flip a light switch at home?"
- Pair them to discuss, then share answers before matching.

3. Interactive Matching Game

- Create physical cards labeled with the words "LED light", "Circuit", and their definitions.

- Have students match the cards in small groups as a warm-up.

4. Guided Discussion Using Visuals

- Show images of circuits, wires, and lights while pointing to each item.
- Say aloud: “This is the circuit—it’s the whole path for electricity!”
- Reinforce with repetition and gestures.

5. Use Sentence Starters:

- “I think a circuit is important because...”
- “The light turns on when...”



Builder

Objective: Students will design and construct two types of swings—one standard and one accessible—to demonstrate their understanding of inclusive design in playgrounds.

This supports creativity, empathy, and real-world application of engineering principles related to accessibility and equity.

Materials Needed:

1. For Structure and Frame:

- Popsicle sticks, wooden dowels, or straws (for swing frames)
- Cardboard strips or foam board (for the swing base and supports)
- Paper towel rolls or small tubes (for vertical posts)

2. For Swing Seats:

- Bottle caps or small plastic containers (for regular swing seats)
- Wider cardboard bases or folded paper (for accessible swing platforms)
- Small paper trays or takeout sauce cups (for seats with backrests)

3. For Support & Movement:

- String, yarn, or thin thread (to hang the swings)
- Tape or glue (hot glue if adult-supervised for strength)
- Paper fasteners or push pins (for hinge-like joints)

4. For Accessibility Features:

- Ramps (cut from Canson paper or thin cardboard)
- Strips of paper, rubber bands, or twist ties (to act as seatbelts)
- Clay or soft foam pieces (to cushion and support seating)

5. For Decoration & Labelling:

- Markers, stickers, or colored paper (to decorate and label)
- Small flags or signs for “Accessible Swing” vs “Regular Swing”

6. Base for Playground Setup:

- Cardboard base (A4 or A3 sheet size depending on workspace)

Activity Instructions:

1. Review the Goal:

- Remind students they are building two swings: one typical and one designed for children who use wheelchairs or need extra support.

2. Plan Your Design:

- Ask students to sketch their swings first. What features should the accessible swing have? (e.g., ramp, seatbelt, back support?)

3. Gather Materials:

- Provide cardboard, straws, string, plastic lids, wooden sticks, clay, tape, etc.
- Encourage students to select strong and safe materials, just like real engineers.

4. Build Both Swings:

- Guide students to use materials to construct a regular swing and an inclusive swing.
- Make sure swings are added to their existing playground base or model.

5. Test and Adjust:

- Encourage students to test movement, support, and balance.
- Prompt them to fix or improve anything that doesn't seem strong or safe.

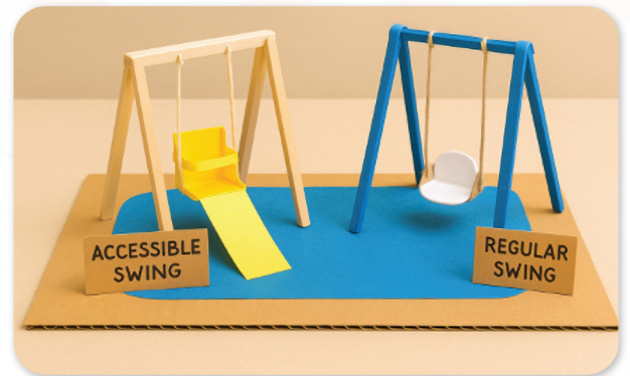
6. Display and Share:

- Students explain how their accessible swing works and how it helps kids with disabilities play safely.

Possible Strategies for Implementation:

- Inspiration First:
 - Show real-world photos/videos of accessible swings (like bucket swings, platform swings, or those with harnesses).
 - Use these to spark ideas.
- Peer Collaboration:
 - Pair students or use small teams to brainstorm and troubleshoot designs together.
- Guided Questions:
 - “How will a child using a wheelchair get on your swing?”
 - “How can you make sure they don't fall off while swinging?”
- Labeling and Reflecting:
 - Have students add small labels to their models: “ramp here,” “seatbelt goes here.”
 - Then ask: “Why did you include this part?”
- UDL Strategy (Universal Design for Learning):

- Offer templates, example models, or visual steps for students who need more support.
- Allow verbal explanations or drawings instead of written reflections for students with writing challenges.
- Safety Check Prompt:
 - Have students perform a “Safety Check” before sharing. Prompt: “Is it strong? Is it balanced? Can all kids use it?”



Evaluate



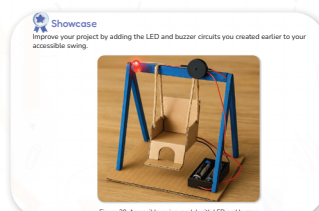
Teacher Role:

- **Facilitator:** Organize a "Light-Up Swing Showcase" where students present their improved accessible swing models with added LED and buzzer circuits.
- **Observer:** Observe how students have integrated the circuits into their swing designs and how effectively the lights and sounds function.
- **Questioner:** Ask students to explain how they connected the circuits and why they chose to place the lights and buzzers where they did to enhance the swing's accessibility and fun.
- **Feedback Provider:** Offer constructive feedback based on the rubric, focusing on the functionality of the circuits and their contribution to the accessible design.
- **Connector:** Help students connect their final projects back to the lessons on circuits, switches, buttons, and inclusive design.

Expected From Students:

- **Project Improvement:** Students will have added functional LED and buzzer circuits to their accessible swing models.
- **Circuit Integration:** Students will demonstrate the successful connection of the LED and buzzer circuits.
- **Explanation of Functionality:** Students will be able to explain how the switches or buttons control the lights and sounds on their swing.
- **Creative Application:** Students will have considered how the lights and sounds enhance the accessibility and enjoyment of the swing for all users.

Parts of the books included.





Showcase

Objective: Students will enhance their accessible swing model by integrating LED lights and a buzzer circuit to increase functionality and safety. This activity encourages application of prior knowledge in electronics and fosters inclusive design thinking through hands-on innovation.

Materials Needed:

- Completed swing models (cardboard, craft sticks, etc.)
- Previously built LED light circuit (light, switch, wire, battery)
- Previously built buzzer circuit (buzzer, push button, wire, battery)
- Additional batteries (if needed)
- Electrical tape or glue dots for mounting
- Scissors
- Markers or labels for decoration
- Optional: Craft decorations for finishing touches

Activity Instructions:

1. Review Purpose:

Remind students that their swings are meant to be accessible. Ask:

“How can lights and sounds help someone know when the swing is in motion or feel safer?”

2. Revisit Circuits:

Ask students to retrieve or reassemble their light and buzzer circuits built earlier. Allow testing time to ensure both circuits still work properly.

3. Plan the Integration:

Instruct students to think about where the LED and buzzer should be placed on the swing to be helpful (e.g., near the seat, on top bar, under seat). Prompt them to sketch their idea on paper if needed.

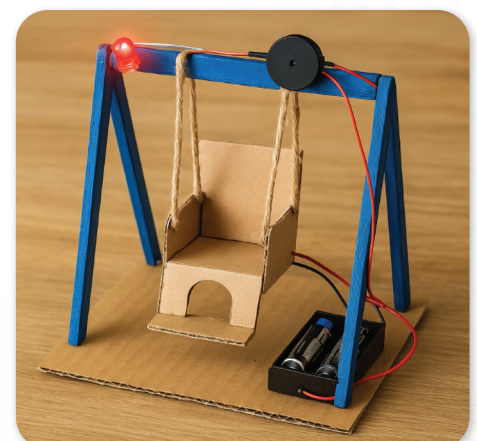
4. Attach Components :

Guide students to:

- Secure their LED light with tape/glue so it is visible.
- Attach the buzzer and switch in a place where it can be easily pressed during motion.
- Ensure the batteries are safely secured but still accessible.

5. Test and Share:

Have students test the swing and demonstrate how the added LED or buzzer helps.

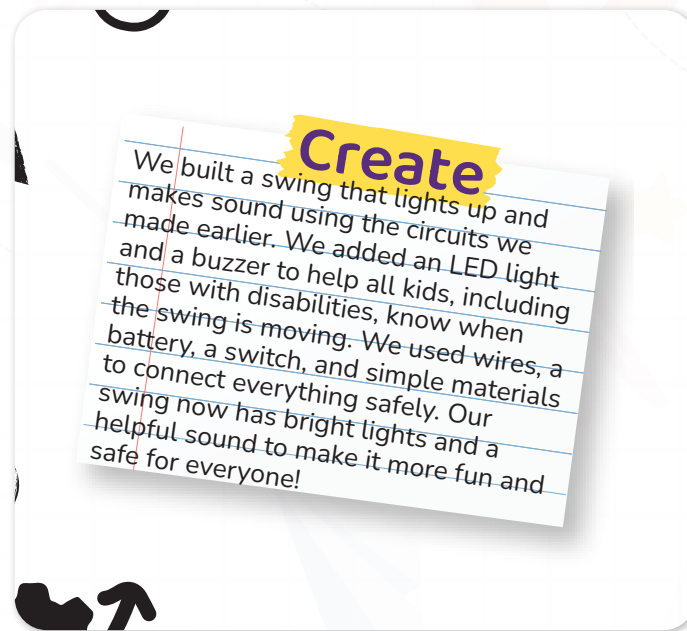


Prompt discussion: “How does this change help kids feel more included or safe?”

Showcase: Rubric for Evaluating Light-Up and Sound Accessible Swings.

Criteria	Beginning (1 Point)	Developing (2 Points)	Proficient (3 Points)
LED Circuit Integration	LED circuit is not added, or it is added but not functioning. Connections are unclear or incorrect.	LED circuit is added, and the LED lights up, but the control (switch) may not be properly integrated or easy to use.	LED circuit is successfully integrated into the accessible swing and functions correctly via a switch. Placement of the LED enhances the design.
Buzzer Circuit Integration	Buzzer circuit is not added, or it is added but not functioning. Connections are unclear or incorrect.	Buzzer circuit is added, and the buzzer makes sound when the button is pressed, but the placement or accessibility of the button may be awkward.	Buzzer circuit is successfully integrated into the accessible swing and functions correctly via a button. Placement of the buzzer and button are thoughtful.
Accessibility Enhancement	The added lights and sounds do not clearly contribute to the swing's accessibility.	The added lights and sounds show some potential to enhance accessibility, but the connection is not well-explained.	The student clearly explains how the lights and sounds (e.g., indicating movement, providing sensory feedback) enhance the accessibility and usability of the swing for all children.
Craftsmanship & Design	The circuits are poorly integrated, appear messy, or detract from the overall model.	The circuits are integrated, but the wiring or placement could be neater or more thoughtfully considered in the design.	The circuits are neatly and thoughtfully integrated into the accessible swing model, enhancing its overall design and functionality.
Explanation of Function	Student struggles to explain how the circuits work or why they were added to the swing.	Student provides a basic explanation of how the lights and sounds work but may not clearly connect them to the swing's accessibility.	Student clearly explains how the LED and buzzer circuits function and articulates how they contribute to making the swing more accessible and enjoyable for everyone.

Don't forget to go back to the EDP page and add these to create part.



Now I Can

Objective: Students will reflect on their learning by identifying how they designed and built an inclusive swing, applied basic circuitry using lights and sound, and used vocabulary related to motion and accessibility. This activity reinforces understanding of STEM concepts, encourages communication, and promotes inclusive design thinking.

Activity Instructions:

- Introduce the Purpose:
 - Begin by explaining that students will reflect on everything they've learned and built in the swing design project.
 - Say: "This checklist helps you think about what you've done and what you've learned. It's okay if you can't check everything—this shows what you're still working on!"
- Student Reflection Time:
 - Allow students quiet time to read each item and check the boxes they feel confident about.
 - Encourage them to reread their journals, look at their swing models, or talk quietly with a partner if they need help remembering.
- Optional Pair Discussion:
 - Have students turn to a partner and say one or two things they are proud of being able to do from the list.
- Class Share-Out (Optional):
 - Invite a few students to share one item they checked and explain what they did.
- Teacher Observation/Documentation:

- o Use this time to circulate and observe which skills students have mastered and which need reinforcement.
- o You may choose to collect the books as formative assessment.

Relation to Standards:

1. Next Generation Science Standards (NGSS):

- **K-2-ETS1-2:** Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function as needed to solve a given problem. (Students are refining their functional and accessible swing model.)
- **K-2-ETS1-3:** Analyze data from tests of two objects designed to solve the same problem to compare the strengths and weaknesses of how each performs. (Students are iteratively improving their design by adding functional elements.)

2. Common Core State Standards (CCSS) - English Language Arts (Speaking and Listening):

- **SL.1.4:** Describe people, places, things, and events with relevant details, expressing ideas and feelings clearly. (Students present and explain their enhanced swing designs.)
- **SL.1.5:** Add drawings or other visual displays to descriptions when appropriate to clarify ideas, thoughts, and feelings. (The physical model serves as a visual display.)

3. Common Core State Standards (CCSS) - Mathematics (Measurement and Data):

- **1.MD.A.1:** Order three objects by length; compare the lengths of two objects indirectly by using a third object. (Students may consider the size and placement of components in their model.)